

The Patriots' Opponents Can't Kick!

Scandal or Good Football?

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Ryan Paganetti spotted a statistic that demands an explanation. From 2001 through 2019, the New England Patriots made 86.25% of their field-goal attempts. Their opponents made only 76.83%. That 9.41-percentage-point gap was *by far* the largest in the NFL. Paganetti provided an explanation: During opposing teams' field-goal attempts, the Gillette Stadium video board sometimes showed a reverse camera angle containing a distracting second, offset set of yellow uprights. Could that visual interference have been costing visiting kickers field goals?

It is a *terrific hypothesis*: interesting, diabolical, and consistent with the Patriots' reputation for pursuing every possible competitive edge. Better yet, it is practical and testable because it makes a clear prediction. The video board stays in Foxborough; the Patriots do not. If the screen caused the effect, New England's field-goal advantage *should shrink sharply* when the team goes on the road.

I will show that most of the gap can be explained by factors that have *nothing to do with the video board and everything to do with New England playing high-quality football*. First, the Patriots attempted shorter—and therefore easier—field goals than their opponents. Second, their kickers, principally Adam Vinatieri and Stephen Gostkowski, were substantially better than the opposing kickers they faced. Third, New England's defenders blocked field goals at an unusually high rate, and blocked kicks are recorded as missed field goals.

Yet one part of the mystery remains. Even after accounting for kick distance, indoor versus outdoor conditions, league-wide improvement over time, and persistent differences among kickers, New England's opponents performed well below expectation. Crucially, however, they underperformed *both in Foxborough and when the Patriots went on the road*. The home deficit is eye-catching, but it is *not substantively or statistically distinguishable from the road deficit*. A somewhat larger unexplained deficit remains at home, so the data do not completely rule out a video-board effect. But they do not establish one either. Once kick difficulty, kicker quality, and blocked kicks enter the story, the possible screen effect becomes considerably smaller—and the statistical evidence for it is not especially persuasive.

The eye-popping number

New England made 533 of 618 attempts. Its opponents made 398 of 518. The raw net advantage was therefore

$$86.25\% - 76.83\% = 9.41 \text{ pp.}$$

Across all 32 teams, the mean net difference was -0.03 pp and the sample standard deviation was 2.61 pp, placing New England 3.61 standard deviations above the team mean.

There is a sharper descriptive comparison. If New England is removed before the reference mean and standard deviation are calculated, the other 31 teams average -0.33 pp with an SD of 2.00 pp. New England's leave-one-out z-score is then

$$z = \frac{9.41 - (-0.33)}{2.00} = 4.87.$$

That is a fair way to ask how far New England sits from the pattern established by everyone else. It also comes with a warning: the Patriots were selected because the statistic looked extreme. This z-score

describes the outlier; it is not a preregistered. We may only be discussing the Patriots because they are the largest value on the list.

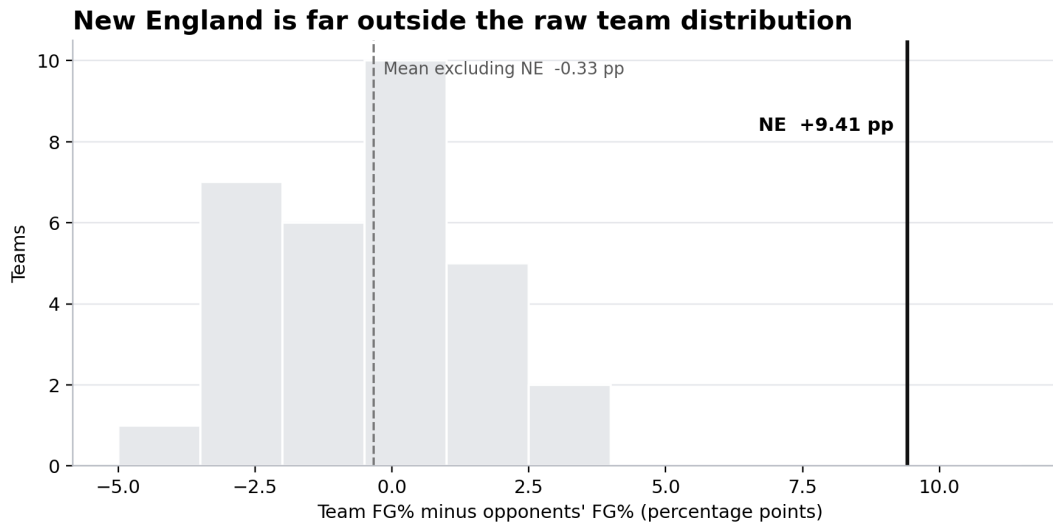


Figure 1: Raw team field-goal differential, 2001–2019. New England is shown separately from the 31-team reference distribution.

The Patriots take the experiment on the road

At home, New England’s kickers beat visiting kickers by 10.02 percentage points. On the road, they beat the home kickers they faced by 8.87 points. The gap barely moves.

That comparison is the key control. Many features of the Patriots travel: players, coaches, protection schemes, rush design, preparation, and kicker quality. The Gillette video board does not. A nearly equal road gap immediately puts pressure on a home-only explanation.

The raw rates still mix together kicks of very different difficulty. New England attempted shorter kicks on average, and the league changed enormously over the sample. A credible expectation must account for the kick, the era, and the kicker.

A better baseline for every kick

To account for this possibly confounding factors, I fit a mixed logistic model to predict the probability of a successful field goal that would adjust for key factors that effect the outcome. The model is fit using every regular-season attempt from 2001–2025:

$$\text{logit}\{P(\text{make})\} = f(\text{season}) + \beta_1 \text{distance} + \beta_2 \text{indoor} + \beta_3 (\text{distance} \times \text{indoor}) + u_{\text{kicker}},$$

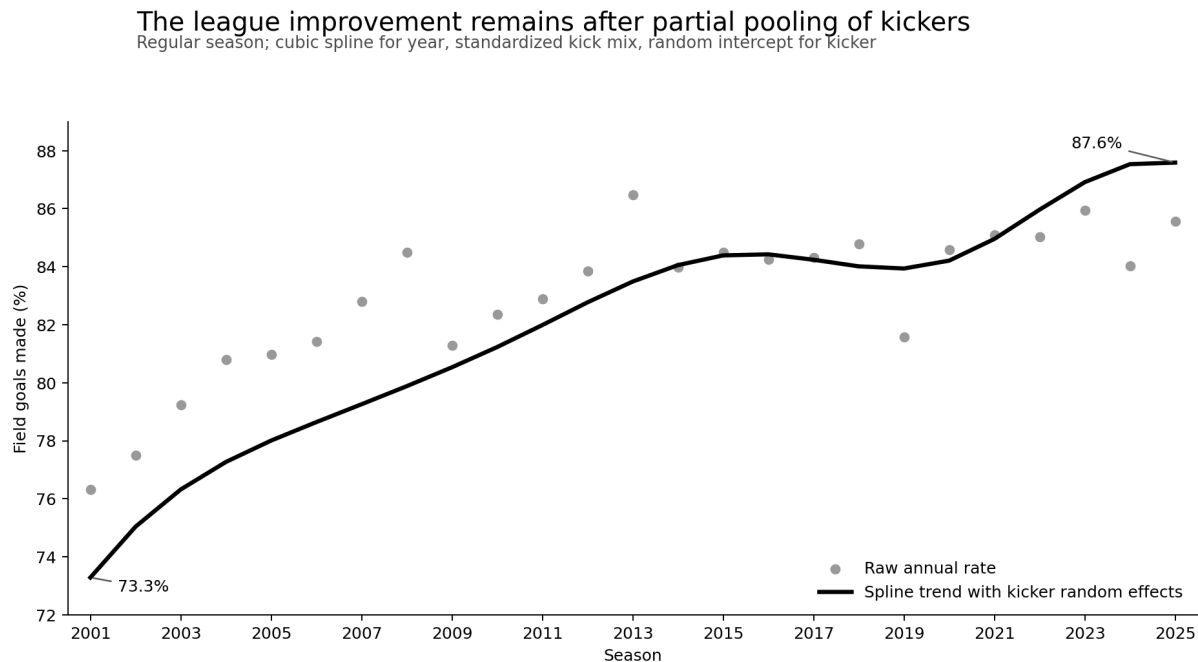
where $f(\text{season})$ is a six-knot cubic spline and $u_{\text{kicker}} \sim N(0, \sigma_k^2)$. Distance and stadium conditions set the difficulty of the attempt. The spline captures the league-wide time trend. The random intercept captures persistent career-level differences among kickers.

The New England analysis uses 18,461 attempts from 2001–2019. The longer 2001–2025 sample—24,773 attempts by 185 kickers—describes the league trend and the scale of kicker heterogeneity.

Kickers improved dramatically

The time curve keeps the mix of kick attributes fixed and varies only the year. For every plotted season, I take all 24,773 observed combinations of distance and indoor or outdoor conditions, set the season to the year of interest, average over the fitted kicker distribution, and then average the predicted probabilities. Each point answers the same question: how would kickers from that season perform against one common representative reference list of kicks?

That reference group has a mean distance of 37.68 yards, an SD of 10.28 yards, and a 25.86% indoor share. Its distance mix is 26.17% below 30 yards, 29.08% from 30–39, 30.10% from 40–49, 14.00% from 50–59, and 0.65% from 60 yards or longer.



The spline curve averages over the fitted kicker distribution and holds distance and indoor/outdoor mix constant.

Figure 2: Standardized league field-goal success under the spline-plus-kicker model. Gray points show the raw annual rates.

Bottom line is the kickers have gotten substantially better. The standardized rate rises from 73.29% in 2001 to 87.59% in 2025—a gain of 14.30 percentage points. The path is not linear. Improvement is steep in the 2000s, slower in the early 2010s, flat around the late 2010s and 2020, then steep again. Models with four through eight spline knots estimate nearly the same kicker standard deviation, from 0.178 to 0.183 log-odds.

How much do kickers differ?

The estimated kicker standard deviation is 0.180 log-odds in the 2001–2025 model. Moving one SD above the league curve multiplies the odds of success by $\exp(0.180) = 1.20$; moving one SD below multiplies them by 0.84. A likelihood-ratio comparison against a model with no kicker variance gives $LR = 18.14$ and a boundary-adjusted $p \approx 0.000010$. Persistent kicker differences are real after distance, stadium conditions, and the league trend are accounted for.

Log odds are hard to understand. The probability effect depends on the baseline. A log-odds shift matters most for a difficult kick near 50% and less for a near-automatic kick.

Table 1: Probability-scale meaning of one kicker standard deviation

Baseline	Kicker at -1 SD	Kicker at $+1$ SD	$+1$ SD change
50%	45.5%	54.5%	+4.49 pp
75%	71.5%	78.2%	+3.22 pp
85%	82.6%	87.2%	+2.15 pp
95%	94.1%	95.8%	+0.79 pp

At an 85% baseline, the central two-SD span runs from 82.56% to 87.15%, or 4.59 percentage points. At a 50% baseline, it spans 8.97 points. “One SD” is therefore not one fixed number of percentage points.

Shrink first, rank second

Kicker effects are partially pooled. A small sample gets pulled hard toward the league-year curve; a long career gets more freedom to separate from it. The approximate median weight placed on a kicker’s own adjusted record is only 1% for ten or fewer attempts, 11% for 26–50 attempts, 25% for 51–100, 38% for 101–250, and 57% above 250. The rest is shrinkage toward the modeled league baseline.

Justin Tucker is the best, hands down, and he provides scale. His fitted career effect is +0.407 log-odds, an odds multiplier of 1.50. Applied to an otherwise 85% kick, that moves the prediction to 89.49%, a gain of 4.49 points. This is a partially pooled career average, not a claim that his ability was fixed from season to season.

Table 2: Selected fitted kicker effects from the 2001–2025 model

Kicker	Attempts	Log-odds effect	Odds multiplier	Effect at 85%
J. Tucker	468	+0.407	1.50	+4.49 pp
J. Hanson	330	+0.213	1.24	+2.52 pp
R. Bironas	279	+0.212	1.24	+2.51 pp
C. Boswell	341	+0.212	1.24	+2.51 pp
J. Kasay	303	+0.207	1.23	+2.45 pp
B. Cundiff	239	−0.205	0.81	−2.80 pp
C. Ryland	90	−0.185	0.83	−2.51 pp
M. Ammendola	35	−0.184	0.83	−2.50 pp
J. Moody	82	−0.180	0.84	−2.45 pp
S. Marler	33	−0.171	0.84	−2.32 pp

The table is an effect-size illustration, not a definitive leaderboard. The model shrinks short careers heavily, and each posterior estimate has meaningful uncertainty which I don’t address here.

Great kickers explain one side of the Patriots gap

The 2001–2019 refit estimates a kicker SD of 0.187 log-odds. Adam Vinatieri’s fitted effect is +0.128 log-odds, an odds multiplier of 1.137. Stephen Gostkowski’s is +0.139, an odds multiplier of 1.149. On an otherwise 85% kick, those effects add about 1.56 and 1.68 percentage points, respectively. The estimates use all of each kicker’s attempts in the 2001–2019 data and then shrink them toward the league curve.

Once those kicker effects are modeled, New England’s own record stops looking mysterious. At home, the Patriots made 85.40% against an expectation of 85.52%, a residual of −0.11 pp. On the road, they

were +1.98 pp above expectation. Overall, they were +0.89 pp above expectation, with a game-clustered $p = .505$.

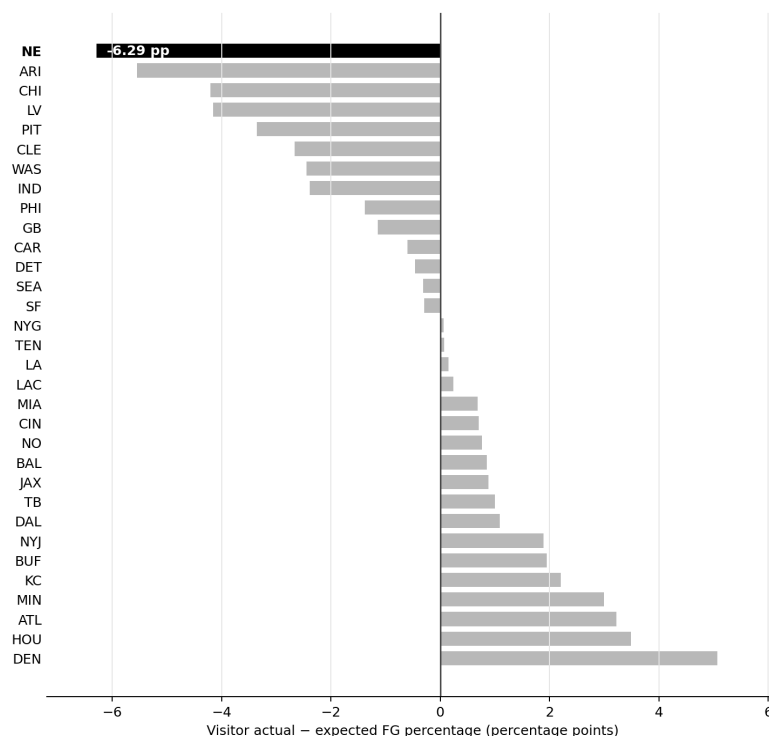
The opponent side gap doesn't disappear when confounders are modeled, but they are not substantially different on the road compared to at home.

Table 3: New England field-goal results under the mixed model, 2001–2019

Context	Attempts	Actual	Expected	Residual	Clustered p	FDR q
NE at home	322	85.40%	85.52%	−0.11 pp	.951	.951
Visitors at NE	260	75.38%	81.67%	−6.29 pp	.00245	.0783
NE on road	296	87.16%	85.19%	+1.98 pp	.294	.952
Hosts vs. NE	258	78.29%	81.61%	−3.31 pp	.153	.668
NE overall	618	86.25%	85.36%	+0.89 pp	.505	.984
Opponents overall	518	76.83%	81.64%	−4.81 pp	.00192	.0615

Visiting kickers at New England finished 6.29 percentage points below their model expectation, the most negative home-opponent residual in the league.

Visiting kickers performed worst relative to expectation at New England
Opponent field goals in each team's home games, 2001–2019 regular seasons



Expected rates control for distance, indoor/outdoor conditions, a spline year trend, and kicker random effects. Negative values indicate visitors made fewer kicks than expected.

Figure 3: Opponent residual in each team's home games. Negative values mean visiting kickers made fewer field goals than expected.

A game-clustered offset-logistic test gives $z = -3.03$ and $p = .00245$. That is strong evidence if New England is the only team under consideration. But I looked at all 32 teams. Benjamini–Hochberg false-discovery-rate control gives $q = .0783$. The Bonferroni-adjusted value is also approximately .0783 because New England has the smallest raw p -value: $32 \times .00245 = .0783$.

The finding is therefore unusual and worth explaining, but it does not clear a 5% family-wide threshold. “Borderline after multiplicity” is less thrilling than “the Patriots broke kicking.” It is also what the evidence says.

The road result is the plot twist

When New England visited, opposing home kickers were still 3.31 percentage points below expectation—the second-most negative value in the NFL. Tampa Bay was slightly lower at -3.39 pp, a difference of just 0.07 pp.

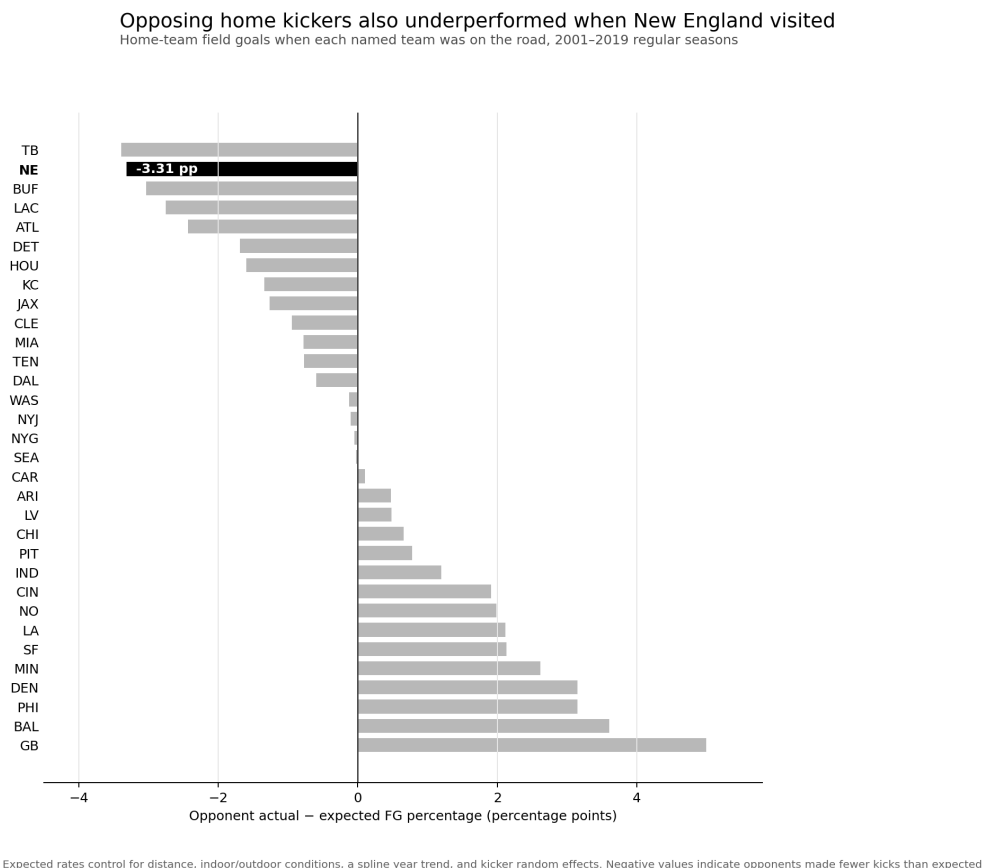


Figure 4: Opponent residual when each named team was on the road. New England ranks second-lowest.

The road residual alone has $p = .153$ and FDR $q = .668$. The more informative test directly compares New England’s opponent effect at home with its opponent effect on the road. The estimated home-versus-road log-odds difference is -0.192 (odds ratio 0.826), with $z = -0.85$ and $p = .395$.

In plain English: the home estimate is lower, but the difference is well within the noise. The data do not show that New England suppresses kickers more at Gillette Stadium than they suppress kickers elsewhere.

Blocks travel, Screens Don’t

Blocked kicks supply a football mechanism that follows the team. Visitors at New England were blocked on 3.85% of attempts. Home teams facing New England were blocked on 3.88%. The two rates are almost identical. New England’s own kickers were blocked on only 0.93% of home attempts and 0.68%

of road attempts.

Table 4: Blocked field goals in New England games, 2001–2019

Context	Blocks	Attempts	Block rate
NE kicks at home	3	322	0.93%
Visitors kick at NE	10	260	3.85%
NE kicks on road	2	296	0.68%
Hosts kick vs. NE	10	258	3.88%

That symmetry fits a traveling special-teams advantage far better than a stadium-only visual effect. It does not prove that blocks explain every miss. It shows that one concrete source of the opponent gap operates in both places.

What the statistic can and cannot support

The Patriots’ raw field-goal gap is real, rare, and worth investigating. But it is not one mystery; it is two. On New England’s side of the ledger, much of the advantage disappears once I account for easier attempts and two exceptional kickers, Adam Vinatieri and Stephen Gostkowski. On the opponent’s side, however, something stubborn remains: kickers facing the Patriots performed substantially below expectation, especially in Foxborough.

The most damaging fact for the video-board theory is also the simplest: *the screen stayed home, but the effect traveled*. Opposing kickers also underperformed when New England went on the road, and the estimated home-road difference is neither large enough nor precise enough to distinguish from statistical noise. Meanwhile, opponents’ kicks were blocked at almost exactly the same elevated rate in New England’s home and road games. That does not prove that blocks explain the entire residual gap, but it points toward a portable special-teams advantage rather than a uniquely Foxborough phenomenon.

None of this proves that the disputed image had no effect. It means only that the aggregate data do not establish one. A cleaner test would identify the individual attempts on which the image appeared and compare them with otherwise similar attempts on which it did not. Until then, the video-board story remains possible, colorful, and unproven. That is the larger analytics lesson: *an extraordinary statistic is an invitation to investigate, not permission to stop*. The best explanation is not merely the one that fits the headline. It is the one that survives the trip to the control group.

Methods note

Team expectations use the fitted random effect for the kicker who attempted each kick. The league time curve instead integrates over the fitted normal kicker distribution. Residuals are tested with the model’s fitted log-odds as an offset in each of the contexts studied (team-locations); standard errors are clustered by game. FDR q-values apply Benjamini–Hochberg adjustment to the 32 team tests within each context.

The kicker intercept is a career-level average. It captures stable differences among kickers but does not let an individual kicker improve or decline over time. The spline captures the common league trend.

Data and acknowledgment

Ryan Paganetti made the motivating observation publicly. Alex Prejean’s related investigation, “How Unusual Was the Patriots’ Kicking Gap?,” is available at byalexprejean.com.

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